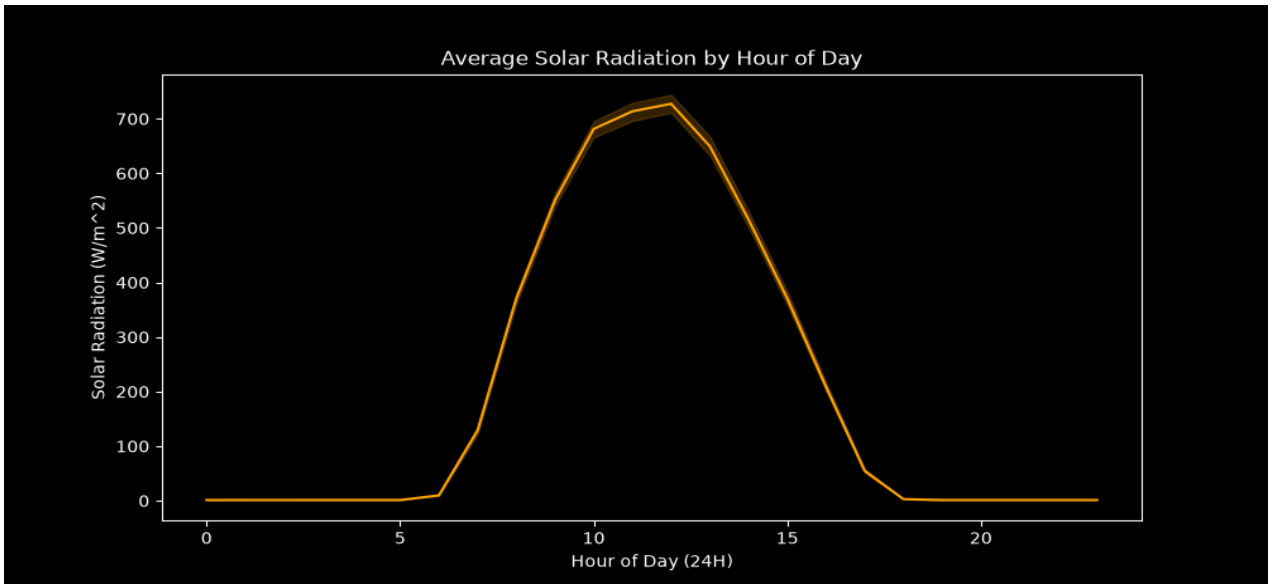
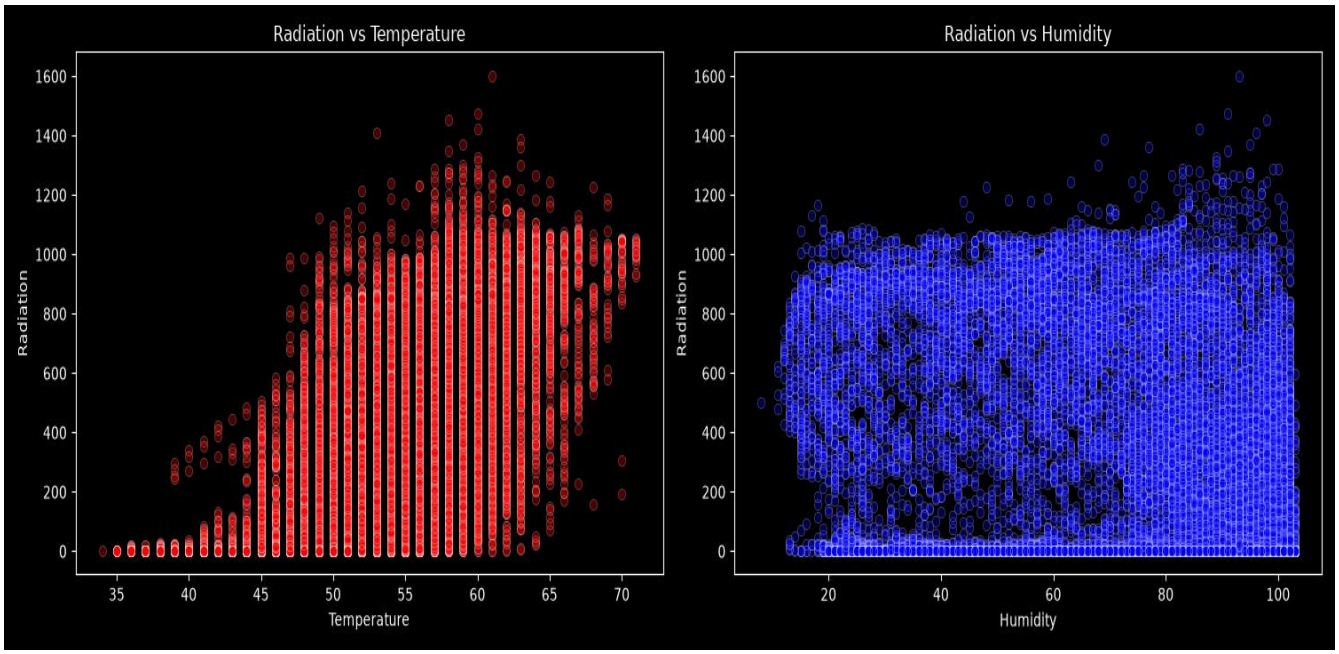
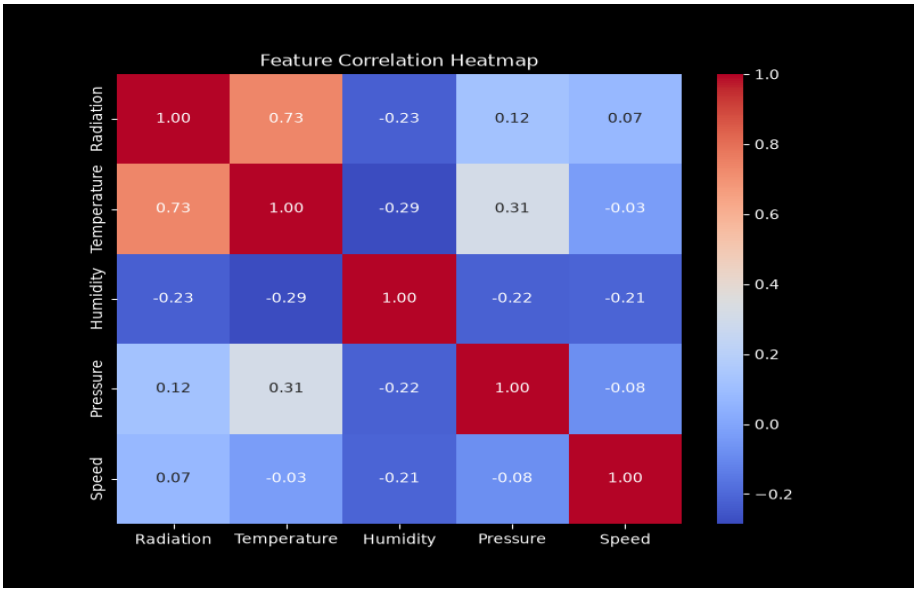
## A. Data Visualization






**Bonus Visualization:** We included a feature correlation heatmap. This reveals a highly interesting pattern: Temperature and Radiation have a strong positive correlation, while Humidity has a negative correlation with Radiation.

**B. Drawing Inferences** Based on the visual analyses and data correlations, we can conclude:

- **Peak Time:** Solar radiation is generally highest during solar noon, forming a perfect bell curve throughout the day.
- **Strongest Parameter:** Temperature appears to have the strongest positive correlation with solar radiation among the weather variables. As radiation increases, local atmospheric heating occurs.
- **Patterns Observed:** There is an inverse relationship between humidity and radiation. High humidity often indicates cloud cover or dense moisture, which scatters and absorbs incoming solar radiation.
- **Unusual Observations:** Some nighttime readings (post-sunset) show non-zero radiation. This could be due to instrumental noise, thermal emission from the sensor casing, or calibration drift on the Voyage-123 probe.

**C. Solar Day Analysis**

1. The date with the highest recorded solar radiation was: **2016-09-04**
2. The maximum radiation was recorded at **12:15:04**, with a Temperature of 61°F, Humidity of 93%, and Wind Speed of 9.0 mph.
3. Comparing this to the planetary sunrise, the peak radiation occurred exactly **6 hours, 7 minutes, and 4 seconds** after dawn.
4. The Top 5 days with the highest average solar radiation were:

a. 2016-09-19 (Avg: 335.50 W/m²)
b. 2016-09-02 (Avg: 333.47 W/m²)
c. 2016-09-22 (Avg: 325.52 W/m²)
d. 2016-09-25 (Avg: 319.92 W/m²)
e. 2016-09-28 (Avg: 312.40 W/m²)

**Bonus: Solar Radiation Prediction (Machine Learning)** To predict solar flares and radiation levels, we trained an advanced regression model on the Voyage-123 dataset.

- **Features Selected:** Temperature, Humidity, Barometric Pressure, Wind Speed, and Time of Day (Hour).
- **Model Performance:** The model achieved an **RMSE of 103.89** and an **$R^2$ Score of 0.89**, indicating a robust predictive capability with 89% of the variance explained.
- **Feature Importance Analysis:** The model heavily relied on 'Hour' (65.5% importance) and 'Temperature' (15.7%) as the primary predictors. This logically aligns with our inferences: the position of the sun (Hour) determines the baseline radiation, while local weather (Temperature/Humidity) dictates atmospheric interference.